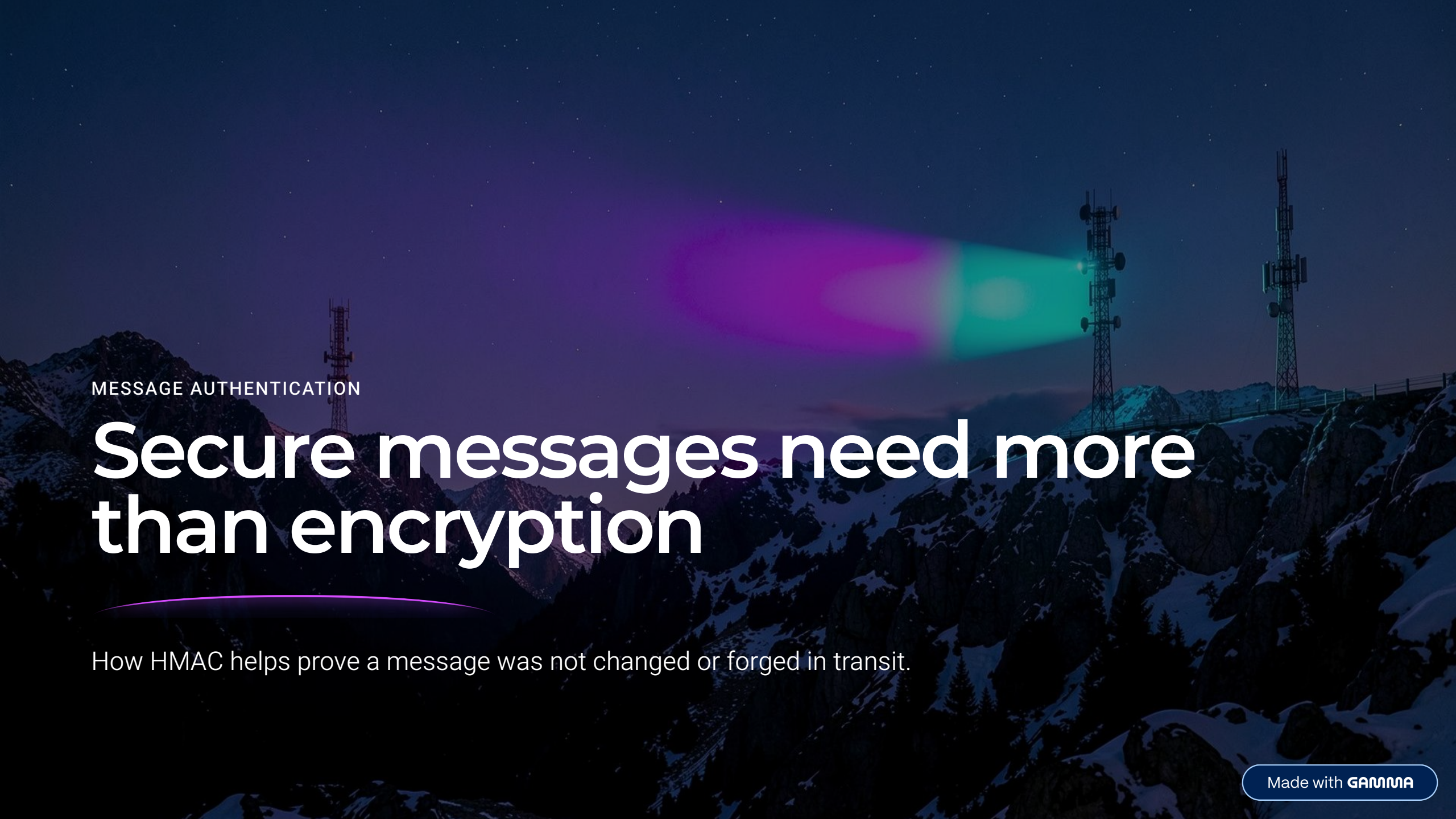


MESSAGE AUTHENTICATION

Secure messages need more than encryption

How HMAC helps prove a message was not changed or forged in transit.

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THE THREAT MODEL

What can go wrong in transit?

A message can arrive looking plausible and still be unsafe to trust.



Changed

Messages can be altered while moving between systems.



Forged

An attacker can send a message that pretends to be from someone else.



Replayed

An old, valid message can be sent again when freshness matters.

The receiver needs evidence about the message, not just a working connection.

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THE CORE QUESTION

Did this come from someone who knows the key?

HMAC answers one question

HMAC provides message integrity and authenticity using a shared secret key.

Both sides must already share the secret.

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THE HANDSHAKE

The shared-secret handshake

The same secret powers creation and checking, but it never travels with the message.



Agree

Sender and receiver agree on a secret key.

Create

The sender combines the key with the message through HMAC.

Check

The receiver recomputes the tag and compares it.

Three steps, one shared secret.

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THE BASIC PATTERN

Message plus tag: the basic pattern

Sender: Create HMAC

Transmit Message + Tag

Receiver: Verify HMAC

The tag is not encryption. It is a tamper-evident seal.



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VERIFY BEFORE TRUSTING

Verification succeeds only when the tag matches

The receiver does not trust the incoming tag by itself. It creates a fresh one and compares the two.

01

Extract

Receiver extracts the message and received tag.

02

Recompute

Receiver computes a fresh tag with the shared key.

03

Compare

Receiver compares tags using a constant-time comparison.

04

Decide

Match means accept; mismatch means reject.

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TAMPER DETECTION

One changed character breaks the match

ORIGINAL

Transfer $100

Tag: 7f2a...c91d

ACCEPT

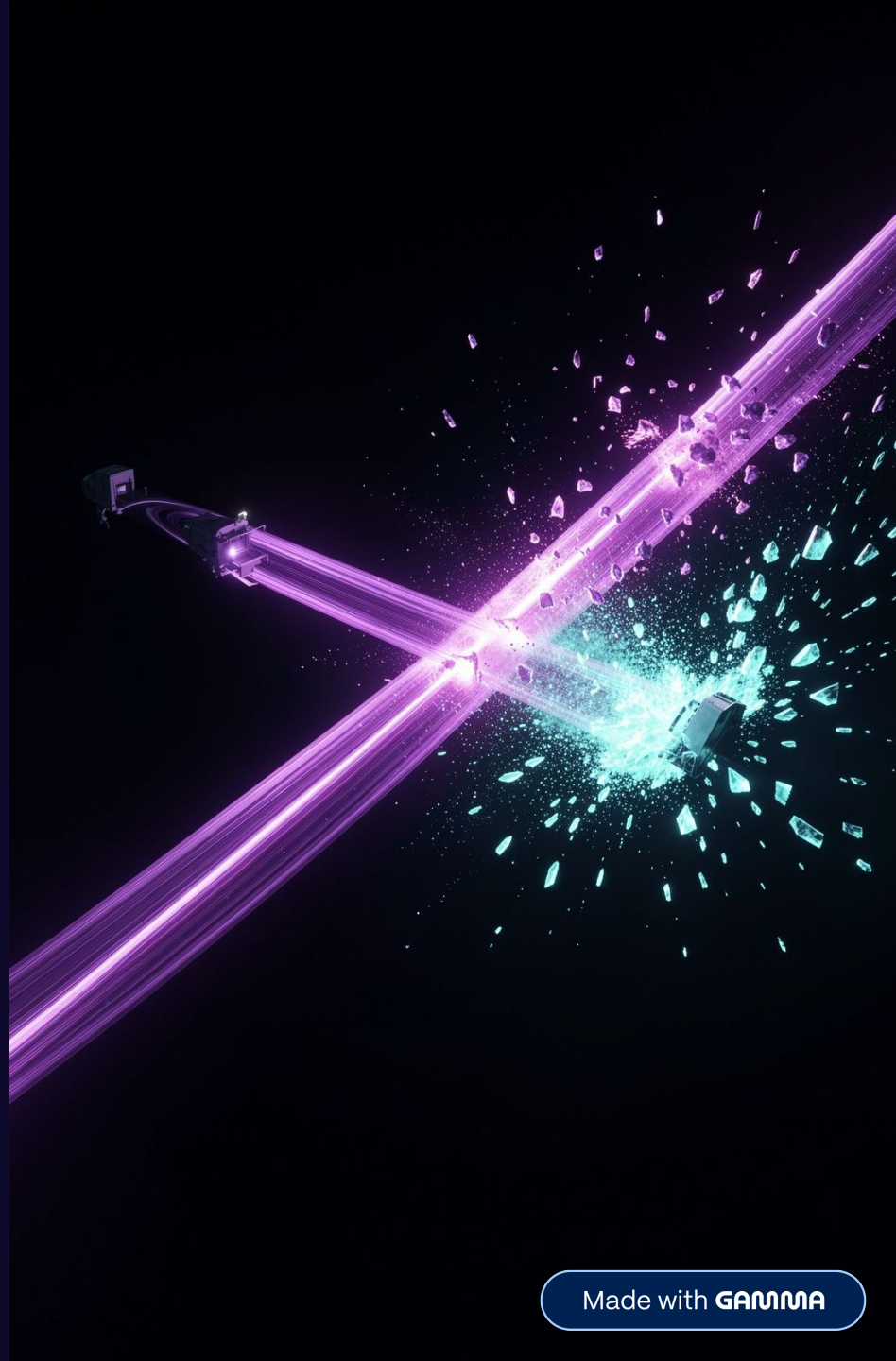
ALTERED

Transfer $900

Tag: 1b84...e03a

REJECT

Changing even a small part of the message produces a different tag, so tampering is detected before the message is trusted.



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TWO DIFFERENT JOBS

HMAC is not encryption

One checks whether content is trustworthy. The other hides what the content says.

HMAC

Integrity and authenticity

Verifies that the message was not changed and came from someone who knows the shared key.

Encryption

Confidentiality

Protects content by hiding it from people who should not read it.

When a system needs both secrecy and tamper detection, use authenticated encryption.

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DON'T ROLL YOUR OWN

Why not hash the key and message yourself?

Cryptography is full of constructions that look reasonable until an attacker finds the edge case.

NAIVE CONSTRUCTION

hash(key ‖ message)

Simple-looking combinations can be unsafe. The details of how the key and message are arranged matter.

STANDARDIZED CONSTRUCTION

HMAC

HMAC is designed to combine a secret key with a cryptographic hash function in a safer, analyzed pattern.

Use a well-tested construction instead of inventing a new one.

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PRACTICAL CHECKLIST

Five habits make HMAC safer



Use a strong, random key

Keep it secret.



Use a modern approved hash

For example, SHA-256.



Separate keys by purpose

Do not reuse one key everywhere.



Compare tags in constant time

Reduce timing leaks.



Add freshness when replay matters

Use a nonce, timestamp, or sequence number.

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